

COSHH RISK ASSESSMENT

Mixed chemical bench preparation

REF CAT-PW-001	ASSESSOR David Kemp	ASSESSED 2026-05-30	NEXT REVIEW 2027-05-30
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Generated May 31, 2026 with CAT — COSHH Assessment Tool

Research Lab

01 OVERVIEW

BUSINESS UNIT

Research Lab

LOCATION

Chemistry prep room

RISK ASSESSOR

David Kemp

ACTIVITY OUTLINE

Small-scale preparation, transfer, storage and cleanup of mixed laboratory chemicals for bench work.

SOP REF NUMBER(S)

SOP-CHEM-015

DATE OF ASSESSMENT

2026-05-30

DATE OF REVIEW

2027-05-30

PERSONS AT RISK

02 TASK HAZARDS

#	HAZARD	RESIDUAL	ACTION
1	Glassware breakage and manual handling during chemical transfer	2	—

03 PROCESS STEPS & CHEMICALS

STEP	ACTIVITY	CHEMICALS	ENGINEERING CONTROLS	PPE
Step 1	Prepare and transfer mixed chemicals	15 chemicals	Fume hood	Gloves, Goggles, Lab coat, Face shield

04 CONTROLS

COSHH screening output

Substance-level screening gives: Approach 1, Approach 2, Approach 3, Approach 4. Highest output: Approach 4 — Specialist advice required, driven by Phenol. This is not a single blanket recommendation; see Appendix C and verify against SDS/task conditions.

Hierarchy controls

Elimination, substitution, reduction and engineering controls are recorded in Appendix D for assessor review.

Administration / PPE / monitoring

Administrative controls, PPE requirements, air monitoring and health surveillance are recorded in Appendix D.

05 STORAGE

Storage and segregation

Storage: Storage groups generated by CAT must be checked against SDS sections 7 and 10 before use. Keep incompatible groups segregated and retain minimum stock.

Incompatibles: Segregate acids/bases, flammables/oxidisers, cyanides/acids, water-reactives/water or aqueous wastes, and toxics from incompatible groups unless SDS confirms compatibility.

06 EMERGENCY RESPONSE AND WASTE

AREA	INSTRUCTION
Spills	Follow SDS Section 6 and the local spill procedure. Only trained staff may tackle small known spills with suitable PPE, ventilation and compatible spill media. Escalate for vapour, toxic, reactive, oxidising, cyanide, drain, fire or uncertainty concerns.
First aid	Use SDS Section 4 as definitive first-aid instruction. Move casualty away if safe, irrigate skin/eyes immediately for corrosive exposure, do not induce vomiting unless medically directed, seek urgent medical advice for toxic/CMR/reactive exposures and send SDS with casualty.
Fire	Raise alarm and evacuate. Only trained staff may use extinguishers for very small incipient fires where SDS Section 5 confirms suitable media. Do not use water on water-reactive materials unless SDS/emergency responders confirm. Provide SDS and chemical inventory to responders.
Waste handling	Collect waste in compatible, labelled, closed containers under secondary containment. Segregate incompatible waste streams; do not dispose to drain unless SDS/local procedure explicitly permits. Use approved hazardous-waste route.
Other	Emergency shower, eyewash, spill kit and SDS access point must be confirmed before starting.

07 BRIEFING & SIGN-OFF

Test user · 2026-05-30

A1 APPENDIX A — CHEMICAL DETAIL

#	CHEMICAL	FORM / QTY	HAZARD STATEMENTS	WEL TWA / STEL	EXPOSURE	GHS
1	Acetone CAS 67-64-1	liquid 250 mL	H225; H319; H336	Check EH40/SDS / — (Manual-EH40)	Up to 30 minutes per use · Weekly · inhalation, skin, eye	 
2	Ethanol CAS 64-17-5	liquid 100 mL	H225; H319	Check EH40/SDS / — (Manual-EH40)	Up to 30 minutes per use · Weekly · inhalation, skin, eye	 
3	Dichloromethane CAS 75-09-2	liquid 50 mL	H315; H319; H335; H351	Check EH40/SDS / — (Manual-EH40)	Up to 30 minutes per use · Weekly · inhalation, skin, eye	 
4	Hydrochloric acid CAS 7647-01-0	liquid 100 mL	H290; H314; H335	Check EH40/SDS / — (Manual-EH40)	Up to 30 minutes per use · Weekly · inhalation, skin, eye	 
5	Sulfuric acid CAS 7664-93-9	liquid 100 mL	H290; H314	Check EH40/SDS / — (Manual-EH40)	Up to 30 minutes per use · Weekly · inhalation, skin, eye	
6	Sodium hydroxide CAS 1310-73-2	solid 100 g	H290; H314	Check EH40/SDS / — (Manual-EH40)	Up to 30 minutes per use · Weekly · inhalation, skin, eye	
7	Nitric acid CAS 7697-37-2	liquid 25 mL	H272; H290; H314	Check EH40/SDS / — (Manual-EH40)	Up to 30 minutes per use · Weekly · inhalation, skin, eye	 
8	Hydrogen peroxide 30% CAS 7722-84-1	liquid 50 mL	H272; H302; H318	Check EH40/SDS / — (Manual-EH40)	Up to 30 minutes per use · Weekly · inhalation, skin, eye	  
9	Sodium cyanide CAS 143-33-9	solid 10 g	H300; H310; H330; H410	Check EH40/SDS / — (Manual-EH40)	Up to 30 minutes per use · Weekly · inhalation, skin, eye	 
10	Phenol CAS 108-95-2	solid 25 g	H301; H311; H314; H331; H341; H373	Check EH40/SDS / — (Manual-EH40)	Up to 30 minutes per use · Weekly · inhalation, skin, eye	  

#	CHEMICAL	FORM / QTY	HAZARD STATEMENTS	WEL TWA / STEL	EXPOSURE	GHS
11	Toluene CAS 108-88-3	liquid 100 mL	H225; H304; H315; H336; H361; H373	Check EH40/SDS / — (Manual-EH40)	Up to 30 minutes per use · Weekly · inhalation, skin, eye	  
12	Formaldehyde solution CAS 50-00-0	liquid 25 mL	H301; H311; H314; H317; H331; H341; H350	Check EH40/SDS / — (Manual-EH40)	Up to 30 minutes per use · Weekly · inhalation, skin, eye	  
13	Potassium permanganate CAS 7722-64-7	solid 20 g	H272; H302; H410	Check EH40/SDS / — (Manual-EH40)	Up to 30 minutes per use · Weekly · inhalation, skin, eye	  
14	Sodium borohydride CAS 16940-66-2	solid 10 g	H260; H301; H314	Check EH40/SDS / — (Manual-EH40)	Up to 30 minutes per use · Weekly · inhalation, skin, eye	  
15	Acetonitrile CAS 75-05-8	liquid 100 mL	H225; H302; H312; H319; H332	Check EH40/SDS / — (Manual-EH40)	Up to 30 minutes per use · Weekly · inhalation, skin, eye	 

B1 APPENDIX B — STORAGE CLASSIFICATION DETAIL

Storage assignments are report evidence from the Storage page. Each row must still be checked against SDS sections 7 and 10 and local storage rules.

#	CHEMICAL	FINAL STORAGE GROUP	SUGGESTED GROUP	HAZARDS	GUIDANCE	SEGREGATION ALERT	CONFIRMATION / REAS
1	Acetone CAS 67-64-1	Flammable Cabinet	Flammable Cabinet	Flammable	Keep away from heat, sparks, ignition sources and oxidizers.	Oxidizers, strong acids/bases, inorganic toxins and reactive materials unless SDS confirms compatibility.	Confirmed Flammable pictogram or flammable H-code trigger.
2	Ethanol CAS 64-17-5	Flammable Cabinet	Flammable Cabinet	Flammable	Keep away from heat, sparks, ignition sources and oxidizers.	Oxidizers, strong acids/bases, inorganic toxins and reactive materials unless SDS confirms compatibility.	Confirmed Flammable pictogram or flammable H-code trigger.
3	Dichloromethane CAS 75-09-2	Toxins Cabinet - organic	Toxins Cabinet - organic	Toxic	Store securely with restricted access. Keep liquids below solids where practicable.	Acids, bases, oxidizers, inorganic toxins and reactive materials unless SDS confirms compatibility.	Confirmed Organic toxic substance from GHS toxicity plus PubChem structural/name data.
4	Hydrochloric acid CAS 7647-01-0	Corrosives Cabinet (Acids)	Corrosives Cabinet (Acids)	Corrosive acid	Store acids separately from bases, cyanides, sulphides and oxidizers.	Bases, cyanides, sulphides, flammables, organic acids and reactive materials unless SDS confirms compatibility.	Confirmed Inorganic/mineral acid from corrosive hazard plus PubChem structural/name data.
5	Sulfuric acid CAS 7664-93-9	Corrosives Cabinet (Acids)	Corrosives Cabinet (Acids)	Corrosive acid	Store acids separately from bases, cyanides, sulphides and oxidizers.	Bases, cyanides, sulphides, flammables, organic acids and reactive materials unless SDS confirms compatibility.	Confirmed Inorganic/mineral acid from corrosive hazard plus PubChem structural/name data.
6	Sodium hydroxide CAS 1310-73-2	Corrosives Cabinet (Bases)	Corrosives Cabinet (Bases)	Corrosive base	Store bases separately from acids and oxidizers.	Acids, flammables, organic toxins and reactive materials unless SDS confirms compatibility.	Confirmed Solid/powder base/alkali from corrosive hazard plus PubChem structural/name data.
7	Nitric acid CAS 7697-37-2	Oxidizers Cabinet	Oxidizers Cabinet	Corrosive acid, Oxidising	Keep away from organic materials, flammables, acids and reducing agents.	Flammables, organic materials, organic acids, organic toxins and reactive materials unless SDS confirms compatibility.	Confirmed Oxidising pictogram, oxidising H-code or peroxide trigger.
8	Hydrogen peroxide 30% CAS 7722-84-1	Oxidizers Cabinet	Oxidizers Cabinet	Corrosive, Oxidising	Keep away from organic materials, flammables, acids and reducing agents.	Flammables, organic materials, organic acids, organic toxins and reactive materials unless SDS confirms compatibility.	Confirmed Oxidising pictogram, oxidising H-code or peroxide trigger.

#	CHEMICAL	FINAL STORAGE GROUP	SUGGESTED GROUP	HAZARDS	GUIDANCE	SEGREGATION ALERT	CONFIRMATION / REAS
9	Sodium cyanide CAS 143-33-9	Toxins Cabinet - inorganic	Toxins Cabinet - inorganic	Toxic	Store securely with restricted access and compatible secondary containment.	Flammables, acids, organic acids, organic toxins and reactive materials unless SDS confirms compatibility.	Confirmed Inorganic toxic substance from GHS toxicity plus PubChem structural/name data.
10	Phenol CAS 108-95-2	Toxins Cabinet - organic	Toxins Cabinet - organic	Corrosive, Toxic	Store securely with restricted access. Keep liquids below solids where practicable.	Acids, bases, oxidizers, inorganic toxins and reactive materials unless SDS confirms compatibility.	Confirmed Organic toxic substance from GHS toxicity plus PubChem structural/name data.
11	Toluene CAS 108-88-3	Flammable Cabinet	Flammable Cabinet	Flammable, Toxic	Keep away from heat, sparks, ignition sources and oxidizers.	Oxidizers, strong acids/bases, inorganic toxins and reactive materials unless SDS confirms compatibility.	Confirmed Flammable pictogram or flammable H-code trigger.
12	Formaldehyde solution CAS 50-00-0	Toxins Cabinet - organic	Toxins Cabinet - organic	Corrosive, Toxic	Store securely with restricted access. Keep liquids below solids where practicable.	Acids, bases, oxidizers, inorganic toxins and reactive materials unless SDS confirms compatibility.	Confirmed Organic toxic substance from GHS toxicity plus PubChem structural/name data.
13	Potassium permanganate CAS 7722-64-7	Oxidizers Cabinet	Oxidizers Cabinet	Oxidising	Keep away from organic materials, flammables, acids and reducing agents.	Flammables, organic materials, organic acids, organic toxins and reactive materials unless SDS confirms compatibility.	Confirmed Oxidising pictogram, oxidising H-code or peroxide trigger.
14	Sodium borohydride CAS 16940-66-2	Reactive Materials Cabinet	Reactive Materials Cabinet	Flammable solid, Corrosive, Water reactive, Toxic	Keep dry, tightly closed, and away from water, acids and incompatible materials.	Water, acids, oxidizers and incompatible aqueous waste streams unless SDS confirms compatibility.	Confirmed H260/H261 water-reactive trigger.
15	Acetonitrile CAS 75-05-8	Flammable Cabinet	Flammable Cabinet	Flammable	Keep away from heat, sparks, ignition sources and oxidizers.	Oxidizers, strong acids/bases, inorganic toxins and reactive materials unless SDS confirms compatibility.	Confirmed Flammable pictogram or flammable H-code trigger.

C1 APPENDIX C — COSHH ESSENTIALS SCREENING

Approaches present

Approach 1, Approach 2, Approach 3, Approach 4

Highest screening output

Approach 4 — Specialist advice required
Driven by Phenol. Hazard group E. H341

Assessor confirmation

This is a substance-level screening output, not one blanket control recommendation. Verify each chemical against SDS, task, route, quantity, duration, WELs and local conditions.

Per-substance breakdown

Substance	Group	H-codes	Scale	Band	EP	Approach
Acetone	A	—	small	high (volatility)	EP2	1
Ethanol	A	—	small	high (volatility)	EP2	1
Dichloromethane	D	H351	small	high (volatility)	EP2	3
Hydrochloric acid	C	H314, H335	small	medium (volatility)	EP2	2
Sulfuric acid	C	H314	small	low (volatility)	EP1	1
Sodium hydroxide	C	H314	small	medium (dustiness)	EP1	1
Nitric acid	C	H314	small	medium (volatility)	EP2	2
Hydrogen peroxide 30%	C	H318	small	low (volatility)	EP1	1
Sodium cyanide	D	H300, H310, H330	small	high (dustiness)	EP2	3
Phenol	E	H341	small	medium (dustiness)	EP1	4
drives controls Toluene	D	H361	small	medium (volatility)	EP2	3
Formaldehyde solution	E	H341, H350	small	high (volatility)	EP2	4
drives controls Potassium permanganate	B	H302	small	medium (dustiness)	EP1	1
Sodium borohydride	C	H301, H314	small	medium (dustiness)	EP1	1
Acetonitrile	B	H302, H312, H332	small	high (volatility)	EP2	1

* assumed value used because input was missing or unparseable.

Caveats & assumptions

Group E hazard (H341) — COSHH Essentials directs to specialist advice (Approach 4). Do not rely on the banded screening.

H304 aspiration hazard: COSHH Essentials does not fully cover aspiration risk — consider design controls against accidental swallowing/inhalation of mists.

Group E hazard (H341, H350) — COSHH Essentials directs to specialist advice (Approach 4). Do not rely on the banded screening.

COSHH Essentials is a screening tool. A competent risk assessor must verify the screening output against the SDS, task, exposure route, quantity, duration, WELs and local conditions, and may impose stricter

D1 APPENDIX D — CONTROL MEASURES DETAIL

CONTROL AREA	RECORDED ASSESSMENT CONTENT
Elimination	Each chemical was reviewed. Remove any substance not required by the SOP before work starts.
Substitution	Substitution must be confirmed against the method and SDS. Use lower hazard alternatives where validated and approved.
Reduction	Use only the minimum working quantity at the bench; return stock containers immediately after dispensing.
Engineering controls	Carry out open handling in a tested fume hood/LEV. Use closed containers and secondary containment for transfer.
Administrative controls	Only trained and authorised personnel may perform the task. Current SDS for all 15 chemicals must be checked before work; reassess before changing chemicals, quantity, concentration or process.
PPE type	Chemical resistant gloves, splash goggles, lab coat, face shield for corrosives/reactives; RPE only if specified by SDS/local assessment.
PPE standard / compatibility	Confirm glove breakthrough/compatibility, eye/face protection and any RPE standard against SDS and local PPE procedure.
Air monitoring	Review EH40/WEL information and SDS exposure limits. Air monitoring is required where exposure cannot be demonstrated as adequately controlled by containment/LEV or after process/control changes.
Health surveillance	Refer to Occupational Health where SDS, local procedure, sensitiser/CMR classification or potential exposure indicates health surveillance or medical surveillance is required.

E1 APPENDIX E — REFERENCE LEGEND

Hazard group	
Group A	Lower health hazard band: H304, H315, H319, H336 and similar lower-toxicity effects.
Group B	Harmful / STOT category 2 band: H302, H312, H332, H371.
Group C	Toxic, corrosive, serious eye damage, sensitising skin, respiratory irritation or STOT category 1/2: H301, H311, H314, H317, H318, H331, H335, H370, H373.
Group D	Fatal acute toxicity, suspected carcinogen / reproductive toxicant, lactation hazard or repeated-exposure organ damage: H300, H310, H330, H351, H360, H361, H362, H372.
Group E	Respiratory sensitiser, mutagen or carcinogen category 1 / suspected mutagen: H334, H340, H341, H350. Specialist advice is required (Approach 4).

Exposure predictor (EP) band	
EP1	Lowest exposure potential: small amounts with low/medium dustiness, or low-volatility millilitre-scale liquids.
EP2	Low-to-moderate exposure potential: grams of high-dust solids, kg/tonne of low-dust solids, or millilitres of medium/high-volatility liquids.
EP3	Moderate-to-high exposure potential: litre-scale liquid work or kilogram-scale medium/high-dust solid work.
EP4	Highest exposure potential: cubic-metre-scale high-volatility liquids or tonne-scale medium/high-dust solids.

Control approach	
Approach 1	General ventilation and good working practice.
Approach 2	Engineering control, normally local exhaust ventilation or equivalent capture / control.
Approach 3	Containment or enclosure where small breaches may occur.
Approach 4	Specialist advice required: the banding screen alone is not sufficient.

GHS pictogram legend

